

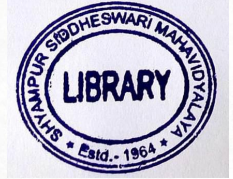
2022

COMPUTER SCIENCE — HONOURS

Paper : CC-8

(Data Communication Networking and Internet Technology)

Full Marks : 50

*The figures in the margin indicate full marks.**Candidates are required to give their answers in their own words as far as practicable.*Answer **question no. 1** and **any four** questions from the rest.1. Answer **any five** questions :

2×5

- ☒ (a) Define : bit rate and baud rate.
- ☒ (b) Why is multiplexing done?
- (c) Why is coaxial cable superior to twisted pair cable?
- (d) What is Minimum Hamming distance?
- ☒ (e) How does a single bit error differ from a burst error?
- ☒ (f) Write down the differences between switch and hub.
- (g) FDM is for analog signals, TDM is for digital signals. Explain why.
- ☒ (h) What are the different functions of network layer of OSI model?

2. ☒ (a) What is transmission impairment? Discuss various types of transmission impairments.

- ☒ (b) State the advantages of FM over AM. Differentiate between circuit switching and packet switching.

1+3+2+4

3. (a) Describe the following encoding techniques with suitable diagrams :

☒ (i) QPSK☒ (ii) QAM☒ (iii) FSK

- ☒ (b) Discuss the advantages of fibre optic cable.

- ☒ (c) Find out the number of links in a mesh topology with n number of devices.

6+2+2

4. (a) Discuss the advantage of two dimensional parity over simple parity. Explain with suitable example.

- ☒ (b) Given a 10 bit sequence 1010011110 and a divisor 1011. Find the CRC.

4+6

Please Turn Over

5. (a) What is channelization? Explain TDMA with example.
(b) Explain how digital information is transmitted over an analog channel. 6+4
6. ~~(a)~~ Discuss the need of ARP and RARP.
~~(b)~~ Why is dynamic routing preferred over static routing algorithm in a network, which changes continuously?
~~(c)~~ Why do we need a DNS system? What is intranet? 4+3+(2+1)
7. ~~(a)~~ State the basic difference between TCP and UDP.
~~(b)~~ Explain the use of SMTP.
~~(c)~~ What is the purpose of transparent bridge? Define bandwidth of a media.
~~(d)~~ Twisted pair cable offers better bandwidth than untwisted pair cable. How? 3+3+2+2
8. ~~(a)~~ Discuss and differentiate between persistent CSMA and non-persistent CSMA.
~~(b)~~ How does Manchester encoding differ from differential Manchester encoding?
~~(c)~~ State the functions of DNS. 4+2+4
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